Supplementary Information

Do pollen-supply constraints underlie floral island syndromes?

This Supplementary Information accompanies the Q1-only empirical Chapter 1 manuscript selected by `config/chapter1_submission_current.json`. It preserves original statistical source identifiers H1–H4 and all numerical decisions; reader-facing ordering has changed, not the models. Results are bound to the active results tables (see results/h1_final_traitwise_t_20261004/README.md and results/geography_20260924/README.md) and the existing deterministic tables in submission/chapter1_current/supplement/Table_S1_data_summary.csv and related supplementary files.

Five current Main display items and their immutable source analyses are listed in FIGURE_CAPTIONS.md and validated as five PDF/SVG pairs in figures/q1_main5/FIGURE_MANIFEST.json. Figure numbering is for the Q1 Main only; source H1–H4 statistical IDs are unchanged. Historical six-source figures remain available for replay, not alternative Main displays.

The Chapter 1 argument has three stages: published experimental pollen limitation (ecological constraint), the original island plant database (seven trait responses **and** the source-H2 selfing-adjusted floral decomposition), and the exploratory exact-species trait–PL bridge. The GloPL samples are not the same sampled island floras; the two datasets cannot identify historical PL mediation. The separate `izu-core` Chapter 2 evolutionary simulations are **not included** in the empirical Chapter 1 analysis or this SI. The geography correction remains post-hoc measurement repair, and H4 remains post-hoc functional triangulation.

S0. Reader-facing sections versus frozen source analyses

Chapter 1 narrative stage	Existing source analysis	Main evidence; what the SI retains
1. Ecological constraint	original H3	GloPL site-level isolation–PL coefficient; details and null sensitivities in S5
2A. Reproductive and floral patterns	original H1	Seven independent atomic traits × four regions; full denominators,

Chapter 1 narrative stage	Existing source analysis	Main evidence; what the SI retains
		pointwise tests and provenance scopes in S3
2B. Selfing-conditioned floral responses	original H2	`F ~ D + selfing_core + area + climate`, treated as conditional statistical decomposition; full direct/All and raw colour–architecture checks in S4
3. Functional compatibility	original H4	Separate exact-species GloPL score fits; species counts, post-hoc status and atomic alternatives in S6

The earlier editorial six-edge D/R/F/P diagram is an index of potential *pairwise associations*, not six equally powered tests, a single joint causal graph, or an alternative source model. Selfing (R) coefficients in original H2b are exploratory readouts; the originally primary H2b estimand was the remaining isolation (D) association after R adjustment.

Adjustment is dataset-specific. The island-trait models control island area and climate PC1–4; the full GloPL analyses control continental-isolation distance, broad geographic context and experimental measurement conditions. Mainland experimental sites have no island area. Neither approach establishes a founder effect, a pollinator decline, or the amount of island-flora change causally explained by PL.

Appendix S1. Corrected geography and analysis scope

S1.1 Corrected island universe

The plant analysis contains 8,264 island units. One GSHHG split component corresponding to continental Eurasia was excluded after the geography audit. The broad H1 analytic union contains 4,379 islands.

The original exposure calculation compared GSHHG island geometry with coarser Natural Earth continental polygons. Within the broad H1 union, 1,113 islands therefore received spurious zero distance. Recalculation against continental coastlines reconstructed from the same GSHHG 2.3.7 high-resolution archive assigned positive distance to all 1,113.

For GloPL, exposure is calculated from study-site coordinates. Of 1,248 unique GloPL sites, 996 lie on seeded continental land and legitimately retain zero distance.

Table S1. Data and geography summary

The deterministic machine-readable version is:

- submission/chapter1_current/supplement/Table_S1_data_summary.csv

Key values are:

Quantity	Corrected value
Global island units	8,264
Broad H1 union	4,379
Accepted angiosperm species	106,295
Possible species × axis cells	318,885
Resolved species × axis cells	222,688
GloPL experimental rows	2,969
GloPL measurement cells	1,408
GloPL unique sites	1,248
GloPL publications	919
True continental GloPL zero-distance sites	996

S1.2 Distance algorithm and validation

Isolation is the minimum coastline-to-coastline separation between minor great-circle arcs on a mean-radius sphere (R = 6371.0088 km). Continental siblings are reconstructed before distance calculation and artificial dateline split edges are omitted. Segment indexing uses triangle-inequality pruning followed by exact arc minima.

Machine-readable receipts:

- results/geography_20260924/spherical_geometry_validation.json
- results/geography_20260924/distance_same_coastline_audit.json
- results/geography_20260924/corrected_universe_exclusion.csv
- results/geography_20260924/formerly_zero_islands_recalculated.csv
- results/geography_20260924/glopl_corrected_site_distances.csv

The numerical validation checks the implementation, not metre-scale accuracy of the coastline source.

Appendix S2. Island floras, trait evidence and redistribution boundary

Observed island floras were assembled from GBIF occurrence records assigned to island units. Trait evidence was normalized to a fixed accepted-species axis and retained source provenance.

The scientific trait database contains 106,295 accepted angiosperm species and three raw evidence axes:

1. flower colour;
2. floral structural complexity;
3. reproductive assurance.

Of 318,885 possible species × axis cells, 222,688 are resolved (69.83%). Species-direct High/Medium evidence forms the Direct-only sensitivity. The primary all-analysis scope additionally admits validated lower-confidence evidence where direct evidence is unavailable. Missing trait information remains missing rather than becoming trait absence.

The full scientific ledger is identified by SHA-256:

a6eef8d731b2730a99e388ddd0683d7ac9b2af30d43f1f9569c857f89f664b2a

The rights-filtered public derivative contains 46,274 redistribution-authorized cells and is archived at Zenodo:

DOI 10.5281/zenodo.22704973

This public derivative is not the complete scientific analysis ledger. Omitted cells are omitted because of redistribution-rights status, not biological missingness or exclusion from the analysis.

Supporting documentation:

- docs/CHAPTER1_DATABASE_RIGHTS_AUDIT.md
- docs/CHAPTER1_DATABASE_PUBLIC_SUBSET.md
- config/chapter1_database_versions/v1.0.0.yml
- submission/chapter1_current/ECOLOGY_LETTERS_DATA_GATE.md

Appendix S3. Final traitwise t inference without multiplicity correction

H1: seven separate trait responses

The primary analysis is broad contemporary island flora with All evidence. Each of seven binary traits is fitted separately in each of four regions by beta-binomial logit regression, adjusting for standardized corrected log isolation, log island area and climate PC1–4. Reproduction, colour and structure are labels, not aggregate scores. Spatial-block sandwich standard errors use a finite-cluster t reference with G−1 degrees of freedom and pointwise 95% intervals. All individual p values are two-sided and unadjusted; no Holm correction is applied. The final rule was user-approved after comparison of alternatives on 4 October 2026 and is retrospective, not preregistered. Individual significance does not establish family-wise significance or formal differences between regional slopes.

WCVP regional-native-compatible flora is the sole active origin sensitivity; Direct-only is a separate evidence-quality sensitivity. Both use the identical model and inference rule. WCVP compatibility retains existing source-native records and upgrades unresolved records only under accepted TDWG-L3 native-range compatibility; introduced records are not overwritten. This establishes regional compatibility rather than exact focal-island nativity. H2–H4 and their previously defined functional covariates are unchanged.

Table S2. Final traitwise H1 results

The active machine-readable H1 table is:

- submission/chapter1_current/supplement/Table_S2_H1_traitwise.csv

It contains all 112 final fits (All/Direct-only × broad/WCVP × four regions × seven traits), their finite-cluster t intervals and pointwise two-sided P values. All 112 fits converged. Earlier joint/vector H1 tables are retained only as historical provenance and are not active submission inference.

H1: recurrent reproductive responses with regional floral differences

All 112 fits converged. Primary All evidence supports 17 of 28 individual associations at nominal two-sided $P < 0.05$, the same set of significant traits as the original poster's normal approximation. Self-compatibility increases with isolation in all four regions ($P = 0.00569$, 0.00671, 0.01812 and 0.01123 in northern mid-latitude, northern high-latitude, tropical and southern extratropical floras, respectively). Selfing mating system increases in the first three regions. Autonomous selfing increases in northern high latitudes and the tropics. Generalized form increases in northern mid-latitudes, the tropics and southern extratropics; actinomorphy increases in

northern high latitudes and the tropics. Shallow/open tubes increase in northern high latitudes but decrease in southern extratropics. Plain colour increases in southern extratropics.

In WCVP All, 10 of 28 individual associations meet the same nominal threshold. Increasing self-compatibility remains supported in all four regions. Northern-high autonomous selfing, actinomorphy and shallow/open tubes remain positive and supported; tropical actinomorphy remains supported. Plain colour increases in both tropical and southern extratropical WCVP floras. Several broad-flora associations, including the southern shallow/open-tube decrease, no longer meet the threshold. This can reflect changed composition, precision and coverage, rather than proving an introduced-species mechanism. Direct-only yields 15/28 supported associations in broad flora and 11/28 in WCVP flora. Complete coefficients, intervals and unadjusted p values are reported in results/h1_final_traitwise_t_20261004/traitwise_results.csv.

Appendix S4. Selfing-syndrome alternatives: source H2 conditional decomposition and raw floral patterns

H2 separates measured reproductive assurance from additional floral response.

The species-level selfing_core score uses compatibility, mating system and autonomous-selfing capacity only. The generalized_accessible score summarizes generalized floral form, actinomorphy and shallow/open tube.

The fitted model family includes:

selfing_core ~ isolation + area + climate

generalized_accessible ~ isolation + selfing_core + area + climate

plain_colour ~ isolation + selfing_core + area + climate

Persistence of an isolation coefficient after adjustment for selfing_core is conditional decomposition, not mediation.

Table S3. Complete H2 conditional decomposition

Active machine-readable finite-cluster tables:

- submission/chapter1_current/supplement/Table_S3a_H2_all_finite_cluster.csv
- submission/chapter1_current/supplement/Table_S3b_H2_direct_finite_cluster.csv

Selfing-adjusted accessibility estimates use t(G−1) P values and BH q values from the final finite-cluster audit:

Region	All-analysis β	SE	P	q	Direct-only β	SE	P	q
Northern mid-latitude	0.02068	0.01281	0.11 03	0.17 64	0.01715	0.01144	0.13 78	0.22 05
Northern high latitude	0.12557	0.03390	0.00 0474	0.00 243	0.11657	0.03716	0.00 2744	0.01 097
Tropical	0.06846	0.02494	0.00 7174	0.01 913	0.05249	0.02616	0.04 751	0.12 67
Southern extra tropical	0.04929	0.03979	0.22 32	0.29 77	0.05779	0.03473	0.10 46	0.20 91

All eight estimates are positive. FDR support is region dependent. In the primary scope it is strongest in northern high latitudes and the tropics. Tropical Direct-only remains nominally positive but is not FDR-supported.

S4.1 Complete-three-component selfing sensitivity

To test whether incomplete measurement of `selfing_core` creates the residual accessibility association, a stricter Direct-only mediator was rebuilt using only species with all three reproductive components observed and informative: self-incompatibility/compatibility, mating system and autonomous-selfing capacity. The complete score was available for 564 species and yielded island-level scores on 2,610 islands before covariate filtering.

Machine-readable results:

- results/h2_complete_selfing_corrected_20261003/h2_complete_selfing_sensitivity.csv
- results/h2_complete_selfing_corrected_20261003/h2_complete_selfing_sensitivity_summary.json

- results/h2_complete_selfing_corrected_20261003/baseline_reconstruction_gate.json

The baseline Direct-only score was independently reconstructed to machine precision before applying the complete-three-component restriction.

Region	Complete-score islands	β isolation	SE	P	q
Northern mid-latitude	1,725	0.02296	0.01107	0.03812	0.05082
Northern high latitude	221	0.08508	0.03433	0.01321	0.03759
Tropical	438	0.05058	0.02153	0.01880	0.03759
Southern extratropical	157	0.03487	0.03165	0.27060	0.27060

All four coefficients remain positive. FDR support persists in northern high latitudes and the tropics, and northern mid-latitudes lie immediately above the 0.05 FDR threshold. This weakens a missing-component explanation for H2 but is not a formal errors-in-variables correction.

S4.2 Raw colour and colour × architecture results

Raw colour, joint colour × architecture and architecture conditional on colour are retained in full rather than compressed into a very large static table.

All-analysis files:

- results/geography_20260924/all/raw_patterns/raw_colour_model_results.csv
- results/geography_20260924/all/raw_patterns/raw_colour_joint_omnibus.csv
- results/geography_20260924/all/raw_patterns/raw_colour_architecture_model_results.csv
- results/geography_20260924/all/raw_patterns/raw_colour_conditioned_architecture_model_results.csv

Direct-only files:

- results/geography_20260924/direct/raw_patterns/raw_colour_model_results.csv

- results/geography_20260924/direct/raw_patterns/raw_colour_joint_omnibus.csv
- results/geography_20260924/direct/raw_patterns/raw_colour_architecture_model_results.csv
- results/geography_20260924/direct/raw_patterns/raw_colour_conditioned_architecture_model_results.csv

These descriptive trait combinations do not identify realized pollinator identity.

Appendix S5. H3 experimental pollen limitation

GloPL pollen limitation is the log response ratio of reproduction after supplemental pollen addition versus natural pollen receipt. Positive values indicate increased reproduction after added pollen.

The corrected analysis contains 2,969 experimental rows, 1,408 publication × coordinate × measurement cells, 1,248 unique sites and 919 publications. Each publication contributes total analysis weight one. Uncertainty is publication-cluster robust.

Table S5. H3 pollen-limitation models

Machine-readable table:

- submission/chapter1_current/supplement/Table_S5_H3_pollen_limitation.csv

Analysis	β distance	SE	Two-sided P	Cells	Publications	Sites
Primary	0.09191	0.03806	0.01594	1,408	919	1,248
Supplemental-only	0.04410	0.04351	0.31134	828	470	736
No-zero-constant	0.09089	0.03865	0.01891	1,375	912	1,238

The primary and no-zero-constant models support a positive isolation association. Supplemental-only remains positive but unsupported. H3 is an isolation-associated pollen-delivery constraint, not direct evidence of declining pollinator abundance or visitation.

Table S5b. Offshore-gradient robustness

Because 996 of 1,248 GloPL sites are true continental zero-distance sites, we tested whether H3 reduces to a mainland/offshore step.

Analysis	β	SE	Two-sided P	Cells	Publications	Sites
Offshore continuous gradient	0.22031	0.09704	0.02319	276	153	252
Mainland vs offshore indicator	0.12743	0.08714	0.14364	1,408	919	1,248

A leave-one-publication-out jackknife of the offshore gradient retained positive estimates in all 153 deletions. The minimum coefficient was 0.17850, the maximum was 0.26930 and the weakest two-sided P value was 0.04892.

Machine-readable results:

- results/h3_offshore_gradient_20261003/h3_offshore_gradient_summary.json
- results/h3_offshore_gradient_20261003/h3_offshore_leave_one_publication.csv

These are post-hoc robustness analyses. They show that the positive H3 association persists within offshore sites and is not captured by a binary mainland/offshore contrast, but they do not establish causation.

Primary source files:

- results/geography_20260924/h3_original_corrected_comparison.json
- results/geography_20260924/h3_corrected_measurement_cells.csv.gz
- results/geography_20260924/h3_corrected_effect_rows.csv.gz

Appendix S6. H4 exact-species functional triangulation

H4 exact-matches the literal H2 species scores to GloPL. Matching uses only case normalization and underscore/space normalization; synonym rescue and genus fallback are not used.

The model is:

pollen limitation ~ H2 score + corrected isolation + geographic context + measurement conditions

Each publication again contributes total weight one and uncertainty is publication-cluster robust.

Table S6a. Exact H2-score bridge

Machine-readable table:

- submission/chapter1_current/supplement/Table_S6a_H4_scores.csv

Trait family / score	Analysis	β trait score	SE	Two-sided P	Species	Publications
Reproductive assurance / selfing_core	Primary	-0.29830	0.10352	0.00417	455	409
Reproductive assurance / selfing_core	No-zero-constant	-0.30078	0.10419	0.00410	453	408
Reproductive assurance / selfing_core	Supplemental-only	-0.09953	0.09398	0.29068	283	241
Accessibility / generalized_accessib	Primary	-0.29566	0.12896	0.02334	143	143

Trait family / score	Analysis	β trait score	SE	Two-sided P	Species	Publications
le						
Accessibility / generalized_accessible	No-zero-constant	-0.30366	0.12991	0.02082	142	142
Accessibility / generalized_accessible	Supplemental-only	-0.29950	0.14562	0.04239	101	98

Negative coefficients indicate that stronger expression of the island-associated H2 score is associated with lower current experimental pollen limitation.

Table S6b. Atomic-trait H4 sensitivities

Primary atomic estimates are reported with finite-publication t inference:

Trait	β trait state	SE	Two-sided P	Species	Publications
Self-compatibility	-0.11885	0.08246	0.1502	499	429
Autonomous selfing	-0.44492	0.08020	4.85 × 10^-8	558	469
Generalized form	-0.18448	0.09174	0.04543	246	246
Actinomorphic symmetry	-0.38076	0.08697	1.47 × 10^-5	582	479

Selfing mating system and shallow/open tube did not pass their parent support gates for atomic H4 promotion.

H4 is explicitly post-hoc functional triangulation. It does not identify historical mediation.

Appendix S7. Inferential boundaries, reproducibility and data availability

The submission distinguishes four inferential levels.

1. Assemblage pattern (H1): contemporary island-flora composition changes with isolation.
2. Conditional decomposition (H2): measured reproductive assurance does not statistically absorb all floral-accessibility response. This is not mediation.
3. Independent ecological pressure (H3): experimental pollen limitation increases with isolation. This is not a direct measure of pollinator abundance or visitation.
4. Functional compatibility (H4): reproductive-assurance and accessibility states associated with isolation are associated with lower current pollen limitation in exact-species overlap. This is post-hoc functional triangulation.

The unobserved historical edge is:

past isolation-associated pollen limitation → selection / sorting / persistence → present trait composition

The current analysis does not distinguish species sorting, differential colonization/persistence and within-lineage evolutionary change.

A separate prospective H4 validation effort stopped at the support gate before outcome unblinding because the prespecified minimum sample/publication requirements were not met. It is a design/support result, not a biological null.

The two earlier predeclared GloPL distance-by-trait moderation families were also replayed after the 24 September geography correction, using the exact archived matched effect rows, trait states, support decisions, models and sensitivities. Before distance replacement, the replay reproduced the frozen estimates to a maximum absolute difference of 3.93 × 10^-14. Correcting the exposure did not rescue either buffering family.

Frozen family / trait	Corrected distance × trait interaction	SE	Two-sided P	One-sided buffering P	Frozen-rule support
Reproductive assurance: self-compatibility	+0.04444	0.07414	0.5489	0.7256	No
Reproductive assurance: autonomous selfing	-0.08329	0.07237	0.2498	0.1249	No
Floral architecture: generalized form	-0.05204	0.09589	0.5873	0.2937	No
Floral architecture: actinomorphy	+0.03413	0.07961	0.6682	0.6659	No

Selfing mating system and shallow/open tube remain support-limited under their frozen preflight gates. Autonomous selfing retains the predicted negative interaction in both frozen measurement sensitivities, but it remains statistically unsupported. Generalized form is negative in the primary and no-zero-constant fits but reverses sign in the supplemental-only sensitivity. Therefore H4 should be interpreted as an association between island-enriched states and lower average current pollen limitation, not evidence that those states flatten the isolation-associated pollen-limitation gradient.

Reproducibility surface:

- results/geography_20260924/corrected_trait_moderation_replay_20261003/
- validation workflow run 37094533233; artifact 11263028661

S7.1 Deterministic supplementary tables

The generated table manifest is:

- submission/chapter1_current/supplement/SUPPLEMENT_TABLES_MANIFEST.json

It records source and output SHA-256 hashes and supports byte-for-byte table rebuilds through:

- scripts/submission/build_chapter1_supplement_tables.py

Generated active tables:

- Table_S1_data_summary.csv
- Table_S2_H1_traitwise.csv
- Table_S3a_H2_all_finite_cluster.csv
- Table_S3b_H2_direct_finite_cluster.csv
- Table_S5_H3_pollen_limitation.csv
- Table_S6a_H4_scores.csv

Atomic H4 sensitivities are reported in the SI text from the corrected atomic source table; earlier H1 joint/vector and domain tables remain historical provenance only.

Table S4 is intentionally represented by the complete corrected raw-pattern file family rather than by one oversized duplicated table.

S7.2 Submission data-policy boundary

The rights-filtered trait derivative is publicly archived at DOI 10.5281/zenodo.22704973. The complete 222,688-cell scientific analysis ledger is not yet wholly redistributable.

Ecology Letters submission remains blocked until either:

1. redistribution rights for the analysis-used trait cells are closed; or
2. the editors explicitly approve a legal/licensing exception and reviewer-access/reconstruction plan.

See:

- submission/chapter1_current/ECOLOGY_LETTERS_DATA_GATE.md
- submission/chapter1_current/DATA_ACCESSIBILITY_DRAFT.md
- submission/chapter1_current/ECOLOGY_LETTERS_DATA_POLICY_INQUIRY.md

S7.3 Reproducibility receipts

- results/geography_20260924/repository_replay_verification.json

- results/geography_20260924/refit_completion.json
- results/geography_20260924/raw_pattern_refit_completion.json
- results/geography_20260924/
  h1_direct_northern_high_convergence_audit.json

This Supplementary Information is descriptive of the current corrected scientific surface; it does not replace permanent data/code archiving.